DEVELOPER INFRASTRUCTURE PITCH DECK

# rapid_ingestor

Kernel-Bypass, Zero-Allocation Telemetry & Log Ingestion Engine

Seastar C++ Reactor Framework

12.4x Vector Throughput



THE CORE INFRASTRUCTURE BOTTLENECK

# Traditional Telemetry Middleware Hits the POSIX & Kernel Wall

### Traditional Middleware (Vector / FluentBit)

- **Kernel Context Switching:** Millions of POSIX socket syscalls daily
- **Thread Lock Contention:** epoll and OS thread scheduling overhead
- **Disk Flushing Stalls:** OS page cache locks during burst writes
- **Write Amplification:** Direct unbatched raw logs choke ClickHouse

### The rapid_ingestor Paradigm

- **User-Space Networking:** Zero kernel context switches on packet ingress
- **Thread-Per-Core Locks:** Isolated core execution without lock sharing
- **Direct DMA Disk Access:** Direct io_uring and O_DIRECT bypassing page cache
- **Native Columnar Formatting:** Zero fragmentation writes for ClickHouse

SEASTAR C++ CORE ARCHITECTURE

# End-to-End Kernel Bypass & Zero-Allocation Pipeline

### Kernel-Bypass Networking

DPDK and AF_XDP ring buffer ingress directly to user memory.

### Thread-Per-Core Reactor

Share-nothing CPU pinning eliminating thread contention locks.

### Direct Async Disk I/O

Linux io_uring and O_DIRECT bypassing OS page cache stalls.

EMPIRICAL PERFORMANCE PROOF

# 12.4x Throughput Advantage Over Vector

**14.37M**
Messages / sec steady ingestion throughput

**>12.4x**
Outperformance vs Vector rust runtime

**2.94 GB/s**
Deterministic flat-line wire bandwidth

Benchmarked on bare metal single core socket parsing with 204.7-byte average log payloads.

ANALYTICAL STORAGE OPTIMIZATION

# Stream-to-Storage Processing Chain

STEP 01

### Ring Ingress

Zero-copy kernel bypass packet capture.

STEP 02

### Zero-Alloc Parse

Native protocol extraction in user memory.

STEP 03

### Column Batch

Direct ClickHouse native block formatting.

STEP 04

### Direct Commit

io_uring flush without page cache locks.

COMMERCIALIZATION STRATEGY

# Open-Core Engine to Enterprise Control Plane

### 1. Open-Source Core

High-performance C++ ingestion engine distributed freely to build developer adoption.

**Bottom-Up Dev Adoption**

### 2. Enterprise Plugins

Monetize corporate identity extensions, LDAP input controls, and compliance tools.

**Inbound Enterprise Pull**

### 3. Managed Cloud

Hosted control plane managing multi-cluster ClickHouse and analytical pipelines.

**SaaS Control Plane**

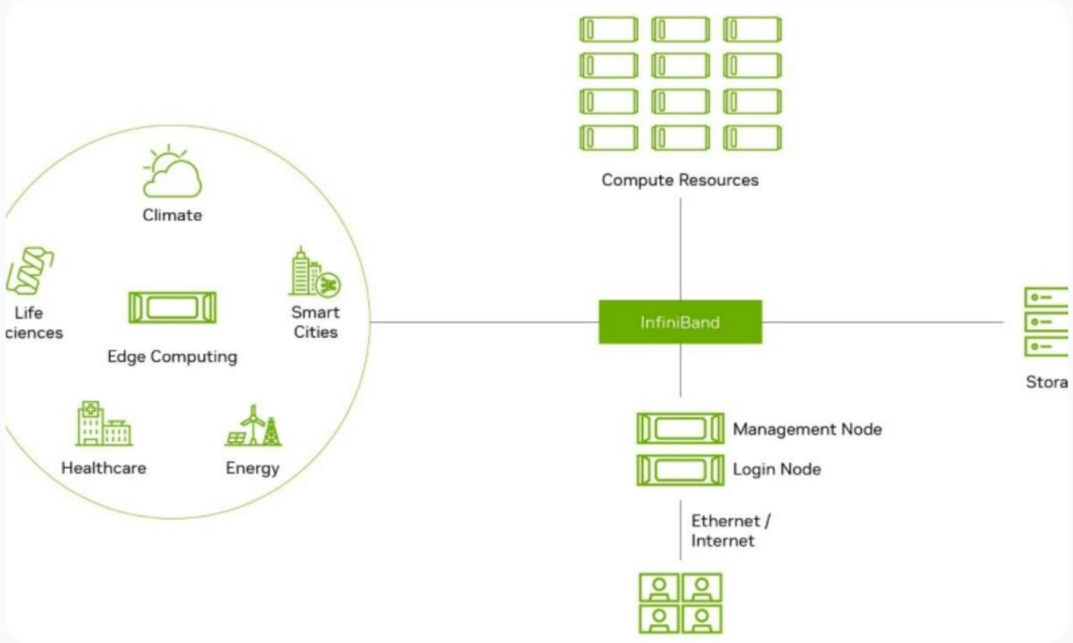
FOUNDER & TECHNICAL OWNERSHIP

# 3 Decades of Low-Level Systems Mastery

## Solo Technical Founder

Built, optimized, and benchmarked rapid_ingestor independently.

- **Deep Systems Expertise:** Extensive track record in C++, Seastar, DPDK, AF_XDP, eBPF, and Go.
- **Datastore Mastery:** Direct experience optimizing ClickHouse and MongoDB storage boundaries.
- **Execution Power:** Demonstrated ability to single-handedly engineer breakthrough infrastructure.



HEAVYBIT PARTNER INVESTMENT ASK

# Building the Next-Gen Telemetry Standard

Raising Day 0 / Pre-Seed capital to convert early inbound pull into enterprise design partnerships and scale core engineering.

MILESTONE 1

### Design Partners

Deliver LDAP enterprise inputs for initial design cohorts.

MILESTONE 2

### ClickHouse Ecosystem

Release native zero-copy ClickHouse block format plugins.

MILESTONE 3

### Team Expansion

Recruit lead core systems engineer and GTM partner.

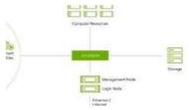
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